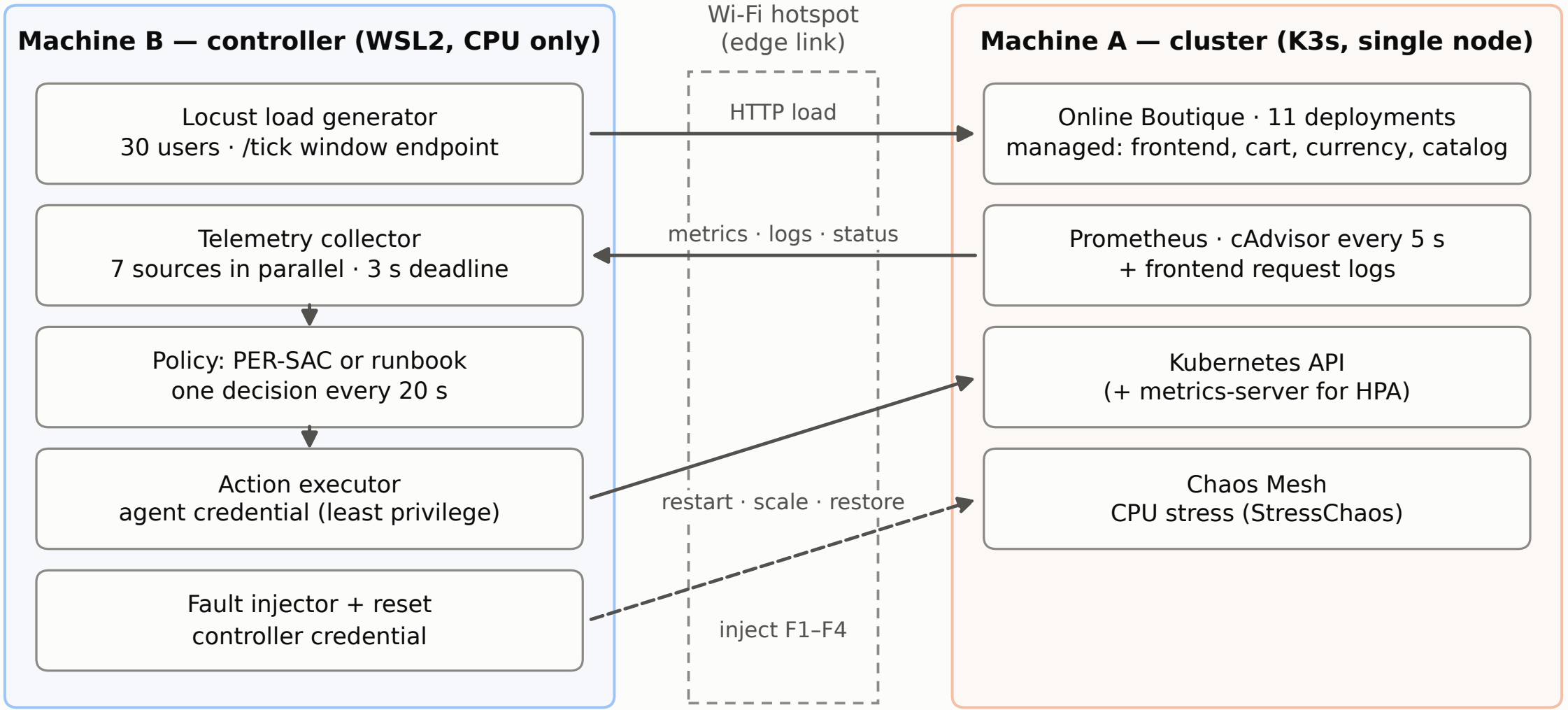
Machine B — controller (WSL2, CPU only)
Locust load generator
30 users · /tick window endpoint
Telemetry collector
7 sources in parallel · 3 s deadline
Policy: PER-SAC or runbook
one decision every 20 s
Action executor
agent credential (least privilege)
Fault injector + reset
controller credential
Wi-Fi hotspot
(edge link)
HTTP load
metrics · logs · status
restart · scale · restore
inject F1–F4
Machine A — cluster (K3s, single node)
Online Boutique · 11 deployments
managed: frontend, cart, currency, catalog
Prometheus · cAdvisor every 5 s
+ frontend request logs
Kubernetes API
(+ metrics-server for HPA)
Chaos Mesh
CPU stress (StressChaos)